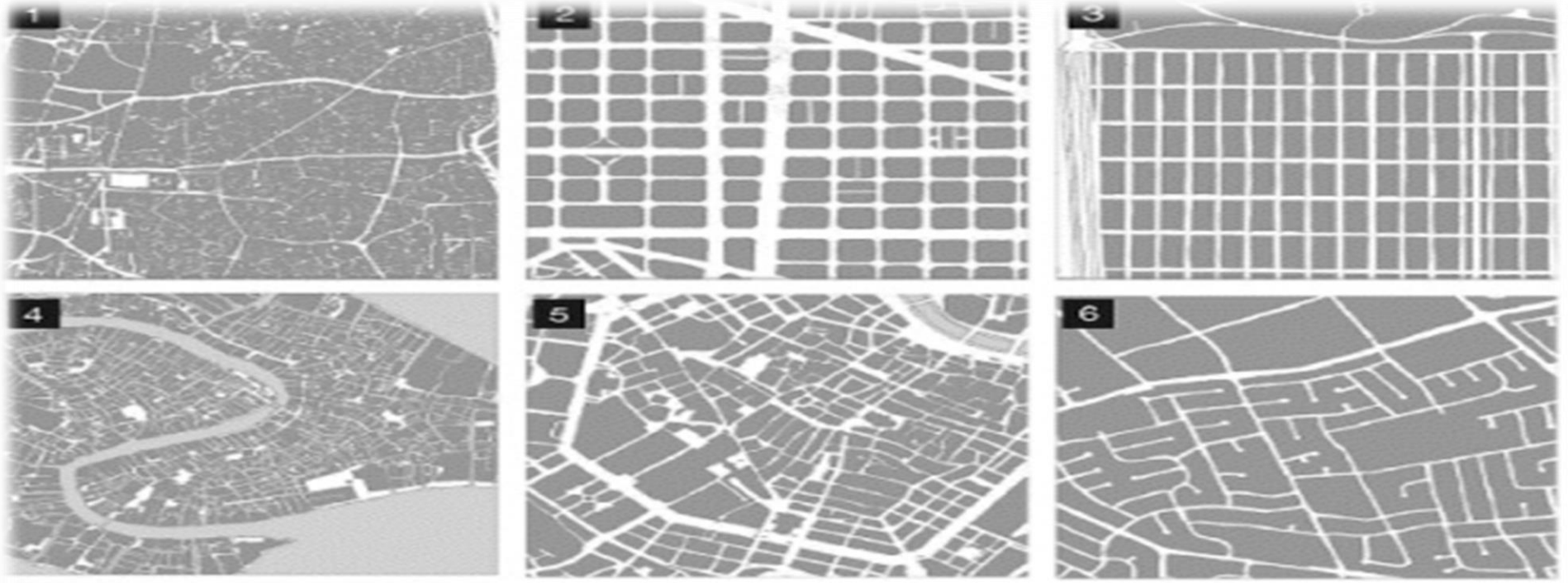


Urban Street Pattern

ARCH 305: Architecture and Town Planning

Last Week

- **Zoning & Zoning Standards**
  - **City/Town level**
  - **Housing Scheme**
  - **House**
  - **Neighborhood Standards**

Contents

- **Street patterns**
- **Types of street patterns**
- **Economy of street patterns**

Introduction

- The street pattern is one of the most important elements of a city plan.
- It forms the framework for the development of the land use pattern
- The location, width, and alignment of a city's streets are all the more important because once established, they are very difficult to change.

Importance of Street Pattern

- The street pattern determines to a large degree
  - the shape, size, and orientation of the individual building sites,
  - the architectural setting of the individual buildings,
  - and beauty of residential, commercial, and industrial neighborhoods,
  - the efficiency of the utility and stormwater drainage facilities,
  - and the safety and effectiveness of the arterial street and
  - mass transit transportation facilities.

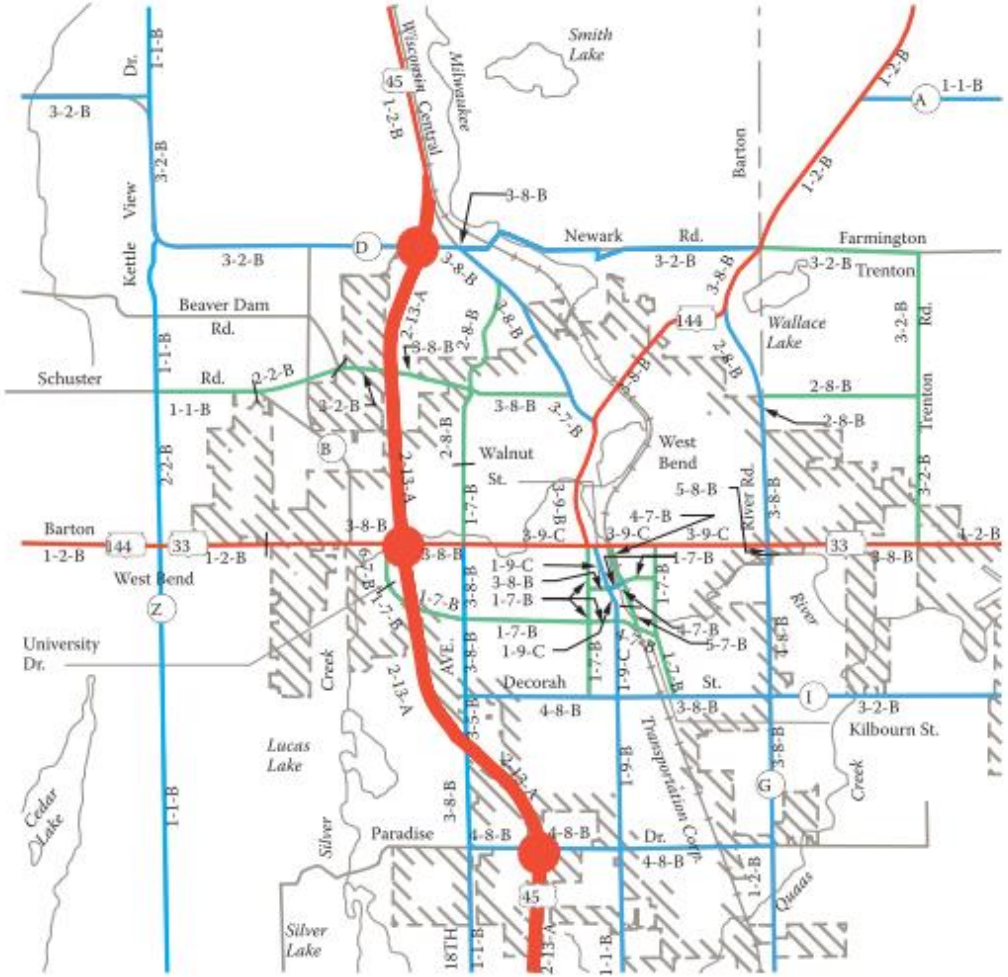
Need of Street Pattern

- Interconnectivity is the important component of neighborhood/city/town.
- Connectivity is achieved through network of roads.
- Interconnections provide different routes for automobile traffic.
- There are a number of different approaches to interconnect streets in neighborhoods and Development Areas.
- Different approached can be combined depending upon the terrain , density and other features of the area.

Functional Classification of Streets

- **Arterials**
  - Freeways
  - Expressways
  - Major Streets
  - Parkways
  - Boulevards
  - By-pass and Ring Roads
- **Collectors**
  - Laneways
- **Land Access Streets**
  - Land Access Streets
  - Alleys
  - Pedestrian pathways
  - Edge streets

Example of Combined Functional and Jurisdictional Arterial Street and Highway System Plan



Legend

Jurisdictional Classification

- Type I arterial (freeway–state trunk highway)
- Type I arterial (state trunk highway)
- Type II arterial (county trunk highway)
- Type III arterial (local trunk highway)
- Freeway–standard arterial interchange

Types of Street Pattern

- The Rectilinear Grid Pattern
- The Diamond Grid Pattern
- The Picturesque Landscape Pattern
- The Stem Pattern
- The Curvilinear Grid Pattern
- The Spider-web or Star Pattern

The Rectilinear Grid Pattern

- **Definition**: The rectilinear grid pattern is a street system providing **maximum road connections** and **some road hierarchy.**
- It represents the classic grid street pattern used in many street systems laid out at the turn of the century.
- This pattern is the **preferred pattern** in the absence of natural features to prevent its use.

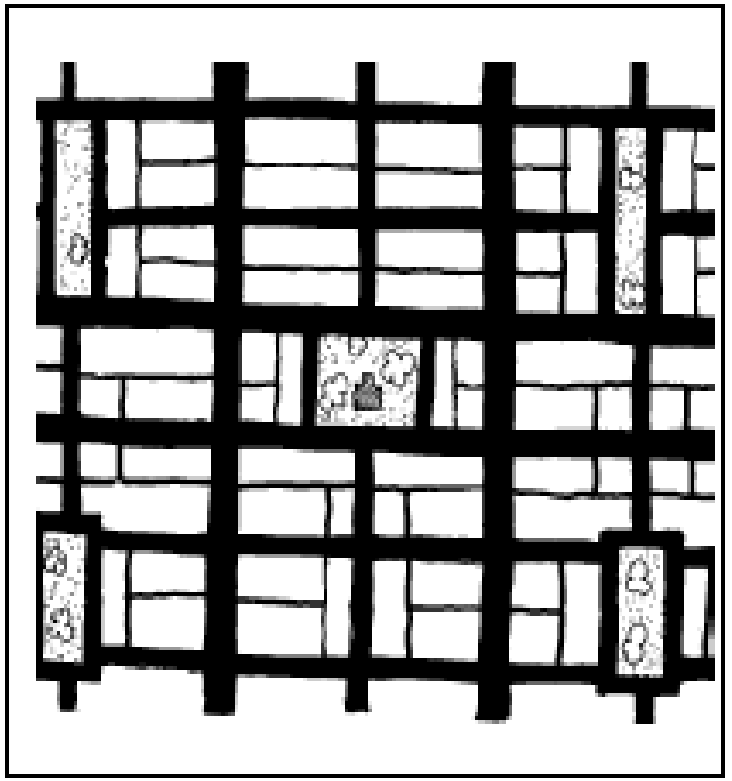


Figure 6:21 Example of a Rectilinear Grid Pattern with residential streets, boulevards, and parks.

The Rectilinear Grid Pattern

Characteristics:

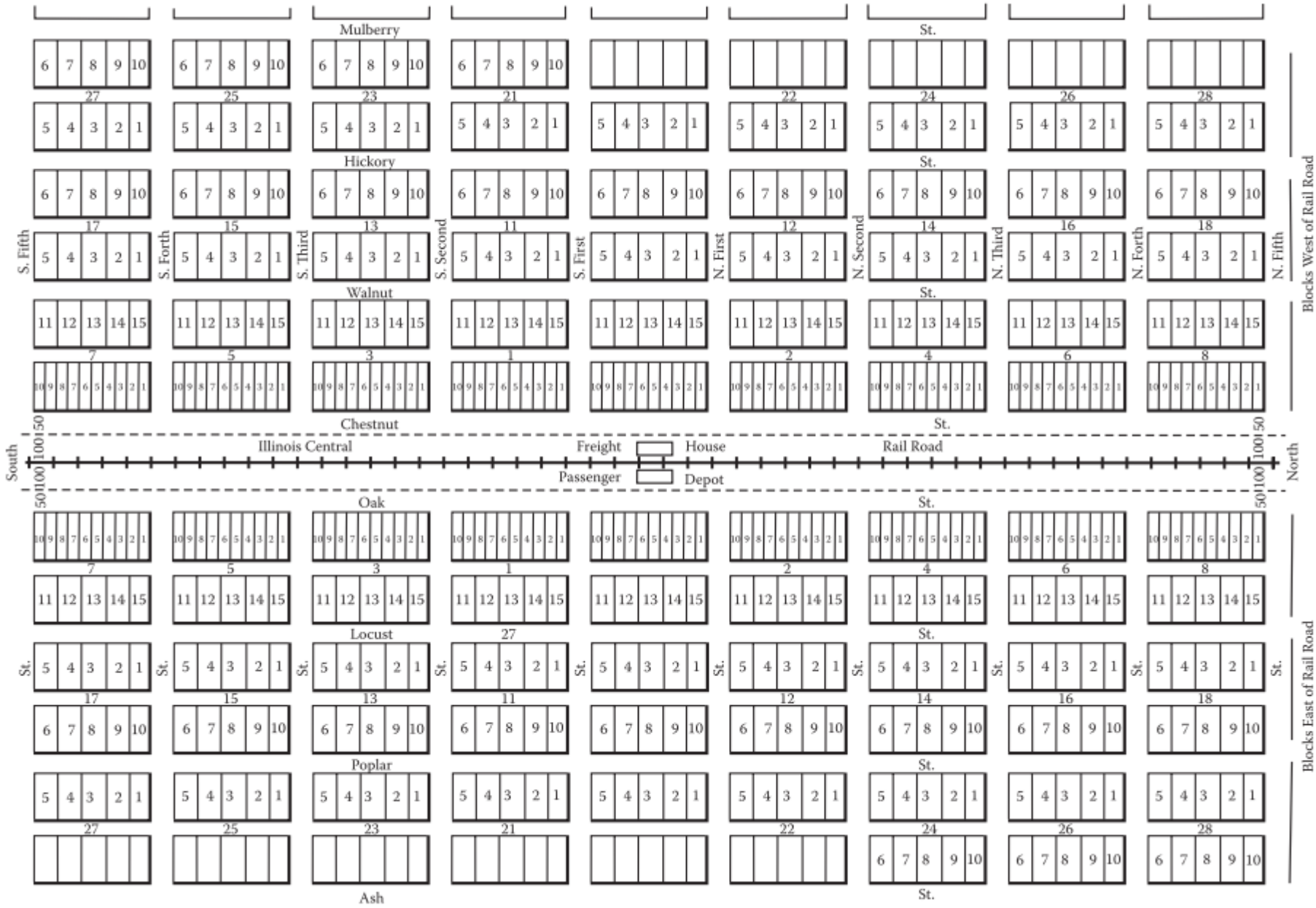
- Provides opportunities for the creation of blocks.
- A hierarchy of thoroughfares can provide opportunities for architectural treatment of buildings at corners.
- It facilitates the use of one-way streets.
- It is easy for people to orient to and is readily house numbered.
- The pattern is easily expandable.

Grid model/Hippodamian plan



Hippodamian plan /Grid Model used in Priene city

Standard town plat from the Illinois Central Railroad: 1851



The Rectilinear Grid Pattern

Characteristics:

- Some streets may terminate at T-intersections, especially in General Areas and Edges.
- it may require excessive amounts of cut and fill for street construction and excessive grading for the development of building sites;
- it may create costly drainage problems;
- Unless it is interrupted periodically, it can be monotonous and is considered by some to be aesthetically unsuited to use in residential areas.
- It may be an un-stabilizing influence since it allows heavy traffic to shift readily from street to street.

The Diamond Grid Pattern

Definition:

- The diamond grid pattern is a grid street system characterized by interconnections at angles.

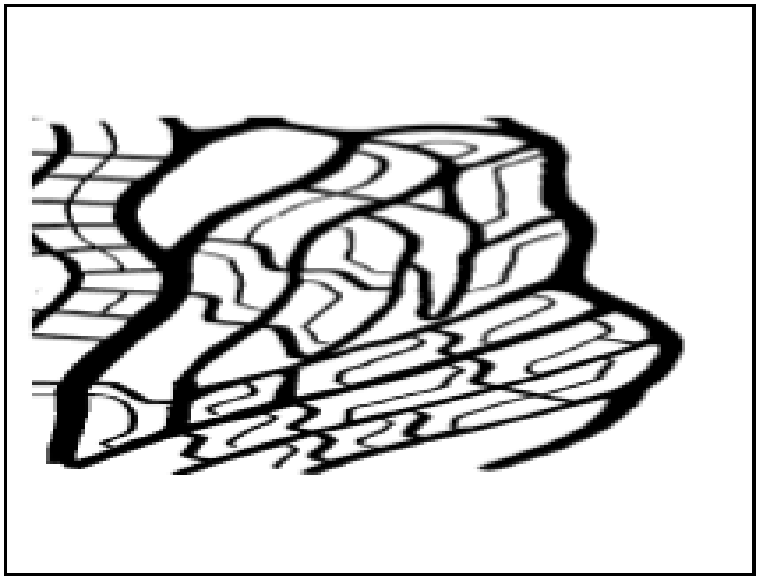
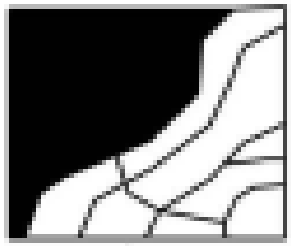


Figure 6:22 Example of Diamond Grid Pattern that provides interconnections and responds to steep terrain.

	Contour Forming	Pronounced terrain relief influences construction of roadways along lines of elevation. Primary streets run parallel to contour lines, with intersecting roads connecting them.

The Diamond Grid Pattern

Characteristics:

- It can work well with steep grades; at extreme grades, a discontinuous pattern may be justified.
- A variety of lot sizes are possible.
- It can produce spectacular and dramatic hill towns.
- It can result in awkward and dangerous intersections.
- Some lots may be awkward in shape and in terrain, making building difficult and expensive.
- Alleys may be difficult to accommodate.
- Siting of buildings relative to streets can be difficult.

The Picturesque Landscape Pattern

Definition:

- Inspired by American landscape architect Frederick Law Olmsteds plan
- Is a loose and warped (partial) grid
- It is able to respond easily to terrain and natural conditions.

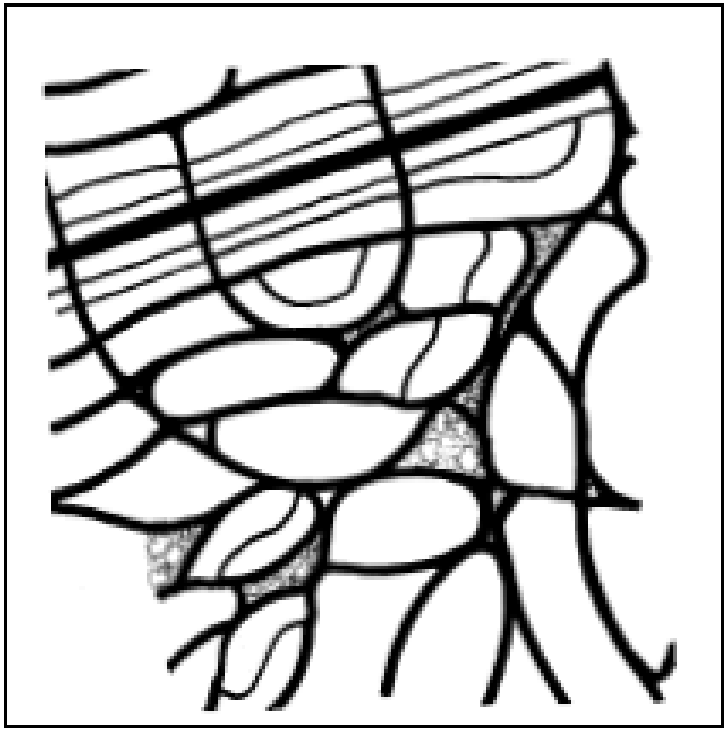


Figure 6:23 Example of picturesque landscape pattern following the terrain and preserving environmentally sensitive areas.

The Picturesque Landscape Pattern

Characteristics:

- It works well with steep grades and is highly responsive to terrain by absorbing environmental features.
- The monotony of the grid is broken by deflected vistas.
- It provides for even dispersal of traffic throughout the network.
- Blocks are not easily created which makes it appropriate only for single-family detached houses.
- It can be very disorienting to motorists, making it easy for them to get lost.
- There is no hierarchy of streets intrinsic to the concept.
- Lot sizes cannot be controlled.

The Curvilinear Grid Pattern

Definition:

- The curvilinear grid pattern is a modified grid that provides **fewer connections** than the Rectilinear and Diamond Grid and more **easily adapts to the terrain**.
- The use of the curvilinear street pattern is generally confined to the relatively smaller areas

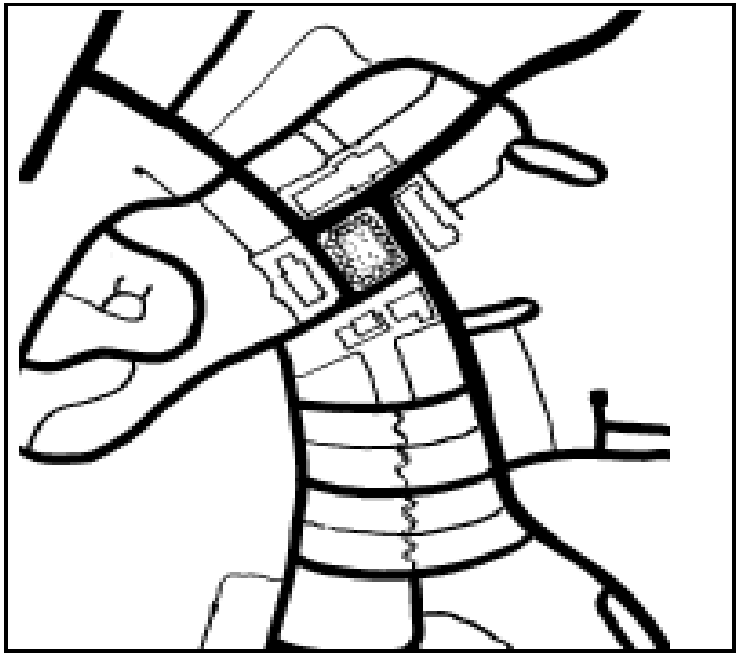


Figure 6:25 Example of the Curvilinear Grid Pattern that runs mostly parallel to topography.

The Curvilinear Grid Pattern

Melbourne



Virginia



Washington Highlands: 1916



The Curvilinear Grid Pattern

Characteristics:

- It can be used with new and existing development patterns.
- A variety of lots can be accommodated and designed.
- It is readily adjusted to the topography of an area,
- minimizing grading and facilitating good drainage.
- It is considered by some to be aesthetically pleasing, curving streets can provide another way to create visual interest.
- providing good architectural settings particularly for residential buildings.
- And it prevents the shift of through traffic from arterials onto collector and land access streets.

The Curvilinear Grid Pattern

- Applied over a larger area, it becomes known as an irregular pattern.
- Its difficult to design and to lay out,
- More difficult for people to orient to,
- and difficult to house number.

The Spiderweb/Radial or Star Pattern

Definition:

- The Spiderweb/radial Pattern is a grid pattern of streets radiating from a center and interconnected mostly at right angles.
- It is a geometrically pure pattern with a central focus.

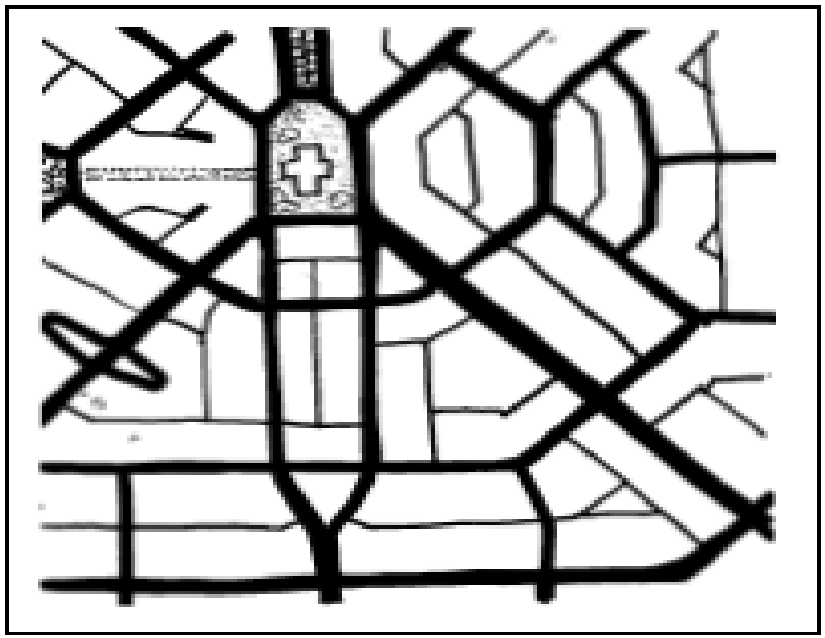
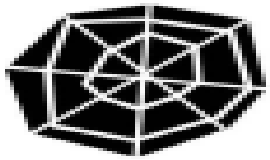
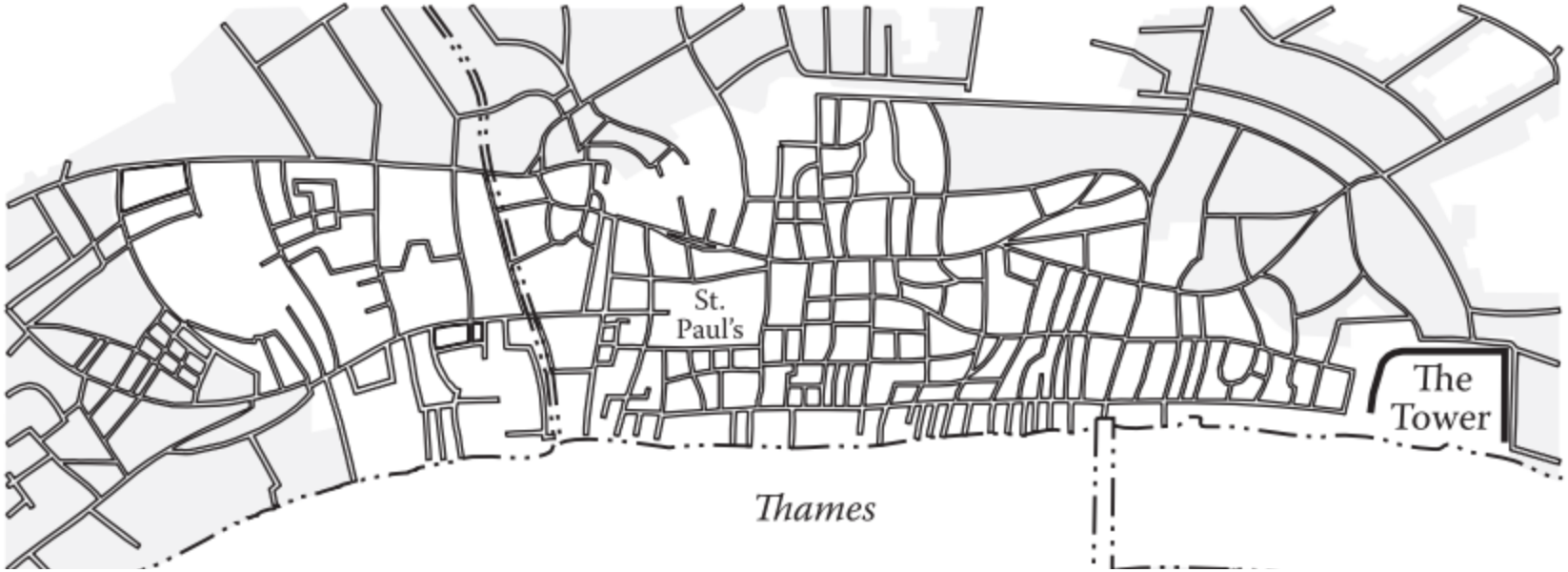


Figure 6:26 Example of Spiderweb Pattern that radiates from a center or monument.

	Radial-Ring	Loops or rings are surrounded by successively larger ones. Usually found in conjunction with larger radial patterns. Radial rings incorporate the elements of both radial and ring/concentric designs.

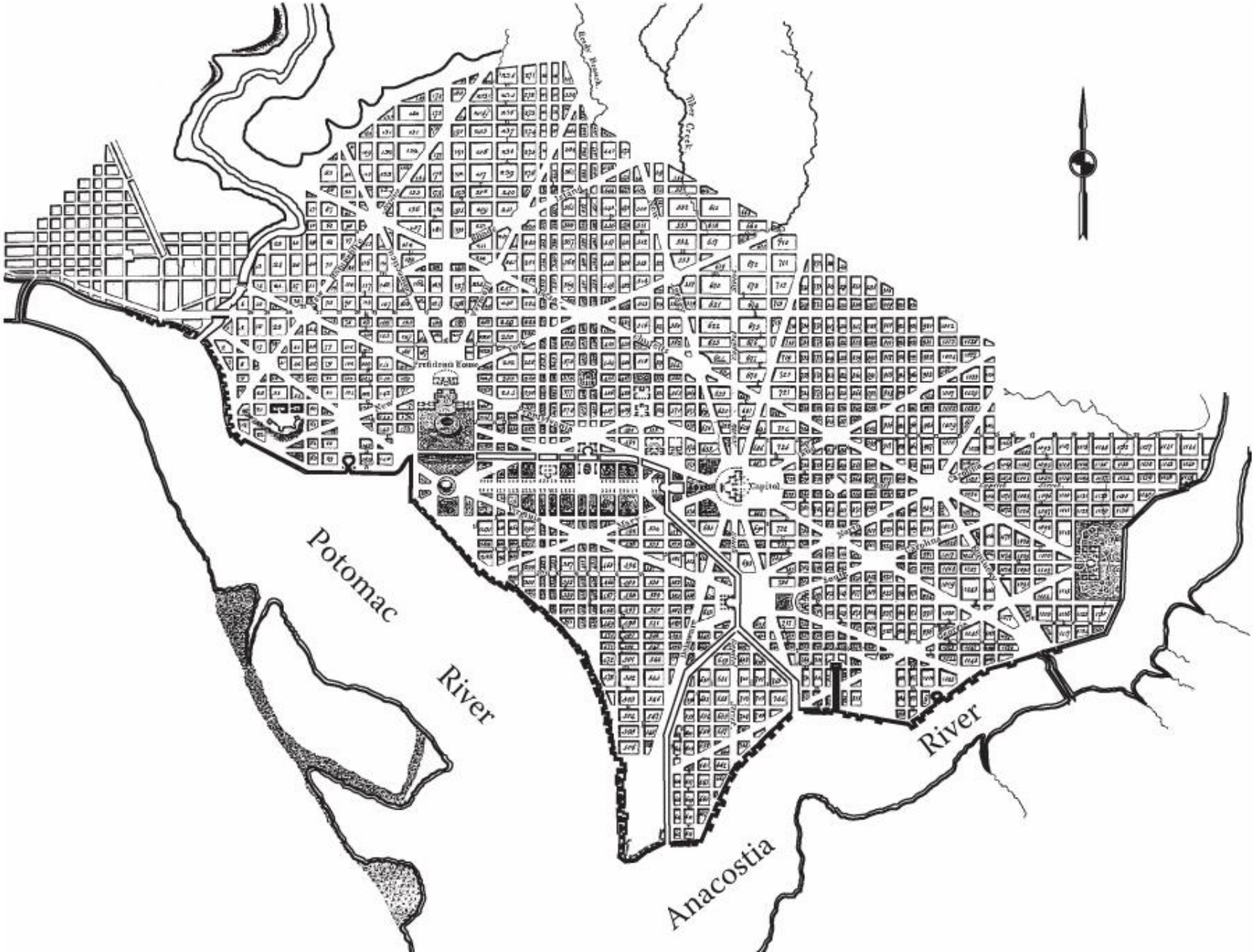
Area Destroyed by Great Fire of London: 1666



Christopher Wren's Plan for Rebuilding the Destroyed Area



Plan for Washington, D.C.: 1791



The Spiderweb or Star Pattern

Characteristics:

It creates formalized central space.

Diagonal streets can create hierarchy for through traffic.

Streets may be positioned to respond to ridges and swales of terrain.

Traffic is dispersed evenly through the network.

The monotony of the grid can be interrupted by deflected vistas.

The street system can be disorienting and difficult for drivers to use.

It requires many special buildings, particularly at the center, in order to achieve spatial definition.

The Stem Pattern

Definition:

- The stem pattern is characterized by a series of cul-de-sac streets feeding onto collector streets and arterials. This pattern is not recommended for standard application in the Development Areas.

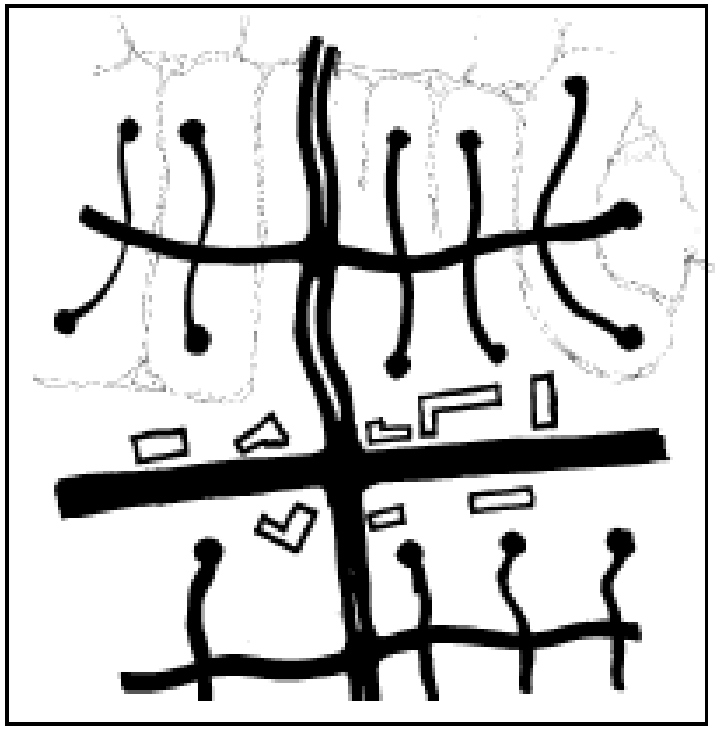
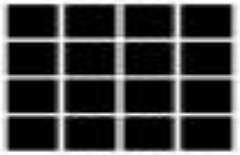


Figure 6:27 Example of the Stem Pattern of cul-de-sacs, collectors, and arterials.

The Stem Pattern

Characteristics:

- It can be responsive to steep terrain.
- It can provide for a maximum number of lots on discontinuous streets.
- It offered a secondary system of pedestrian paths behind each stem of houses to separate pedestrian from vehicular traffic.
- It can contribute to traffic congestion by the absence of an interconnected network.
- A complete separation of land uses and housing types is intrinsic to the pattern.
- High design speeds and wide roadways can lead to excessive speed.

	Rectangular or Chessboard	Streets are grid-like, with parallel streets intersected by perpendicular streets.
	Rayed	Streets that fan out at various angles from a given focal point and through less than 360 degrees.
	Radial	Primary thoroughfares radiate out from a central point. These streets may be extended outward 360 degrees around the central point or within an arc from a point along a natural barrier, such as a coastline.
	Radial-Ring	Loops or rings are surrounded by successively larger ones. Usually found in conjunction with larger radial patterns. Radial rings incorporate the elements of both radial and ring/concentric designs.
	Contour Forming	Pronounced terrain relief influences construction of roadways along lines of elevation. Primary streets run parallel to contour lines, with intersecting roads connecting them.
	Irregular Pattern	Irregular street patterns have been specifically engineered without geometric patterns for aesthetic or functional reasons. An American subdivision with curving streets and cul-de-sacs is an example.
	Combined Pattern	Any combination of the above and is best demonstrated by the development of high rise and business districts in Medieval or pre-Medieval cities.
	Linear Pattern	A primary thoroughfare radiates down the center with buildings on either side. American strip malls and main shopping districts are patterned this way for ease and convenience.

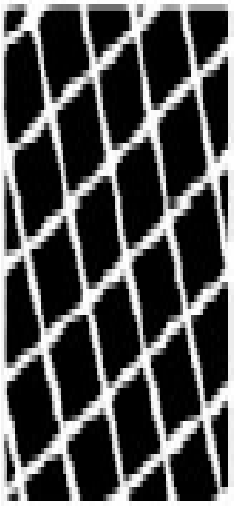
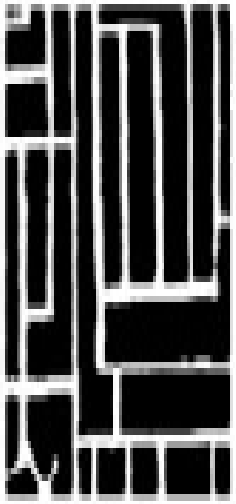
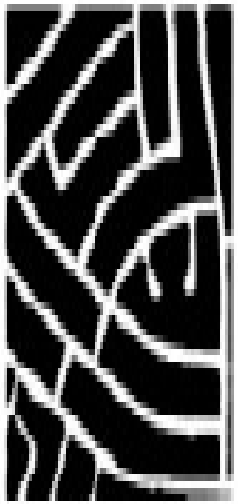
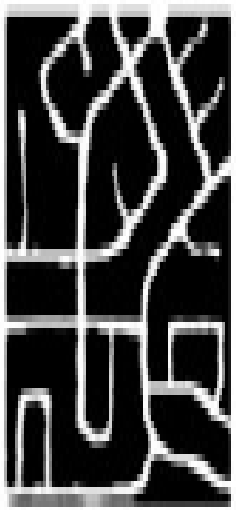
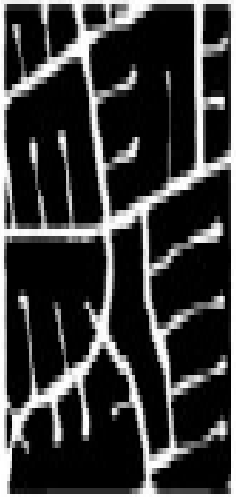
Efficiency and quality

- Efficiency is chiefly the result of combining two standard street ***types—loops and culs-de-sac—with long blocks***.
- Contrary to popular opinion, the curvilinear streets that are typical of conventional **suburban subdivisions are not inefficient**; they reflect an aesthetic preference and have little impact on **land consumption**.

Evolution of street patterns

- **Loop and cul-de-sac street patterns have evolved from 1900 to the present (see Fig.2).**
- Their geometry is adapted to the automobile, excluding traffic at the local street level and permitting good flow at the collector and arterial levels.
- By contrast, the traditional grid patterns that predate the automobile have required major adaptations such as one-way streets and traffic lights in order to achieve good automobile traffic flow.

Evolution of street patterns

	Gridiron (c. 1900)	Fragmented parallel (c. 1950)	Warped parallel (c. 1960)	Loops and lollipops (c. 1970)	Lollipops on a stick (c. 1980)
Street patterns					

Compactness

Difference in block size, building density and road network design



Grand Central Station, New York



Oxford Circus, London



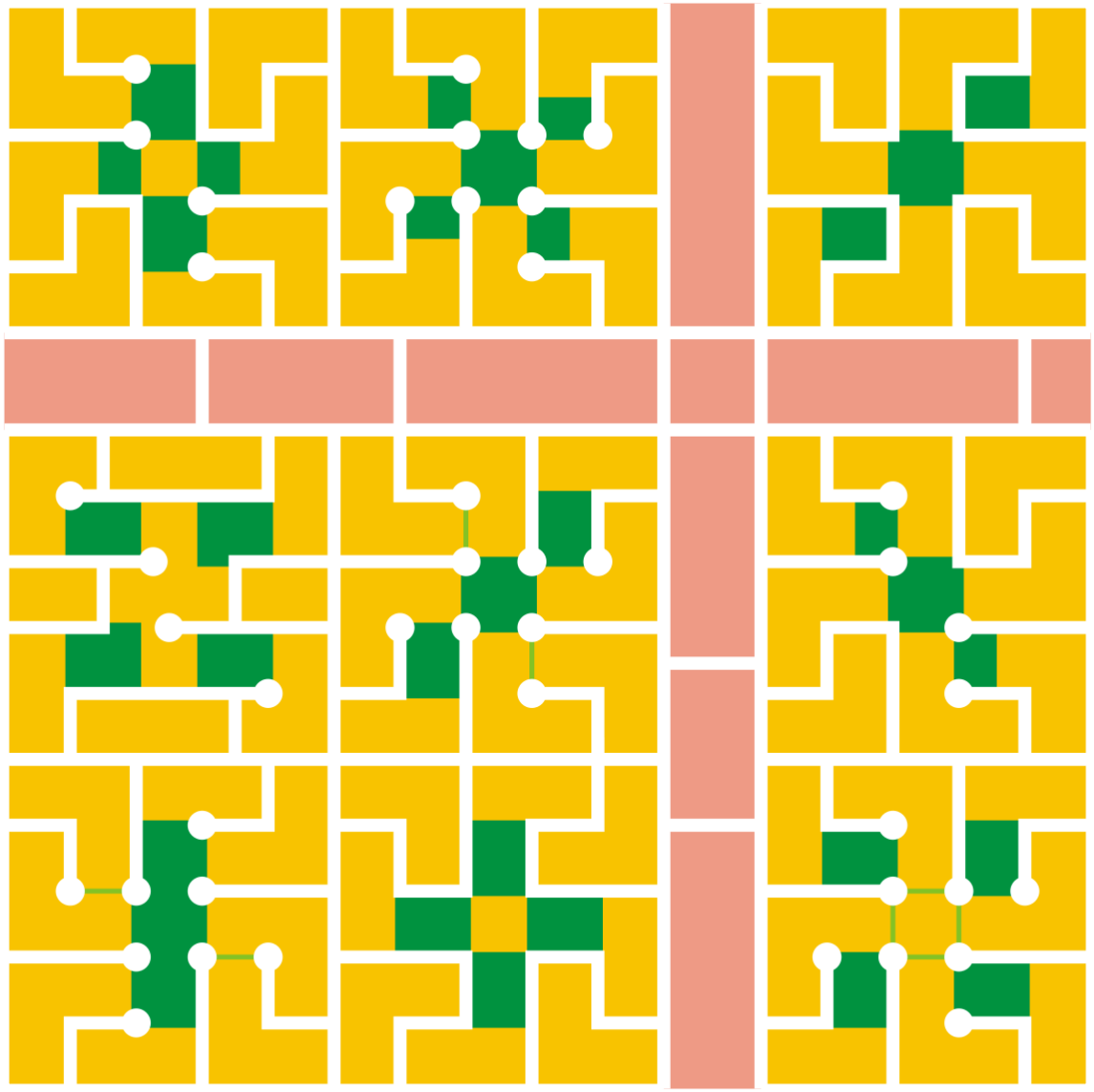
Guomao, Beijing





World Bank

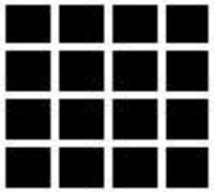
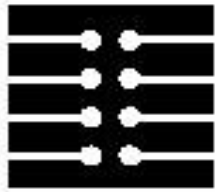
A Game of Cul-de-Sacs



Efficiency and quality

- According to the technical literature on street planning, conventional suburban street layouts consume 16-25 per cent less land than the traditional grids advocated by new urbanism (see Fig. 1.)

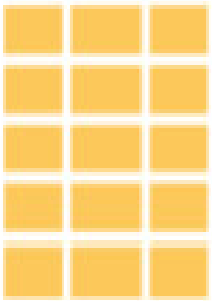
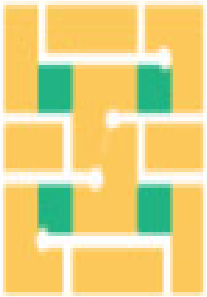
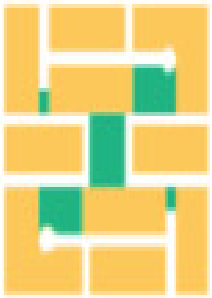
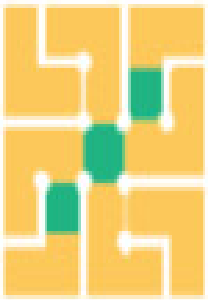
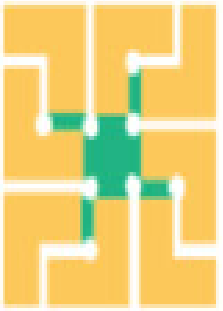
Figure 1: Comparison of area used for streets, among five typical patterns .

	 Square grid (Miletus, Houston, Portland, etc.)	 Oblong grid (most cities with a grid)	 Oblong grid 2 (some cities or in certain areas)	 Loops (Subdivisions - 1950 to now)	 Culs-de-sac (Radburn - 1932 to now)
Percentage of area for streets	36.0%	35.0%	31.4%	27.4%	23.7%
Percentage of buildable area	64.0%	65.0%	68.6%	72.6%	76.3%

Density and Street pattern

- There is no correlation between street pattern type and residential density.
- Density has a strong impact on land usage.
- It is related to the presence of amenities within a district.
- The strongest predictor of residential density is land price, with higher densities occurring where land prices are high.

Figure 7: Buildable area, street area and open space area percentages comparison

	 Square Grid (Most cities)	 VFG A (4 loops, 4 C-D-S)	 VFG-B (4 loops, 2 C-D-S)	 VFG-C (8 C-D-S)	 VFG-D (8 C-D-S)
Per cent of area for streets	35.0%	27.4%	27.4%	23.7%	23.7%
Per cent of buildable area	60.0%	63.6%	64.3%	68.0%	68.0%
Per cent of open space	Required 5.0%	9.0%	8.3%	8.3%	8.3%
Total	100.0%	100%	100%	100%	100%

STREET PATTERN	GRID	CURVILINEAR
Energy consumption	• requires 2.3 lamp standards/acre	• requires 1.2 lamp standards/acre
Safety	• typically will have more intersections which will help slow traffic but also increases potential for vehicle collisions • the increased access to private property is contrary to CPTED principles	• hierarchical street network eliminates through traffic in cul-de-sacs and side streets but increases the volume and speed of vehicles on collector routes
Connectivity	• directness of pedestrian access is facilitated by virtue of street locations and intersecting pattern • sidewalks on both sides of street • Intra and inter-neighbourhood connections is facilitated by the orthogonal geometry of the streets	• connectivity must be consciously built into area structure plans (for inter-neighbourhood connections) and subdivision design (for intra-neighbourhood connections)
Ease of orientation	• the orthogonal geometry of a grid street clarifies location and direction for visitors in the neighbourhood.	• as part of a street hierarchy can alter perceptions of direction but guide movements more directly and intuitively between origins and destinations.
Sociability	• Sidewalks on both sides of the street facilitate pedestrian activity throughout the neighbourhood.	• Cul de sacs and bays allow children to play in the front yard and street without being endangered by fast moving traffic
Housing diversity and affordability	• higher infrastructure costs are incorporated into final house cost • Standard lot sized result in similar consumer for all lots	• the potential of increasing the number of units per acre lowers the per-household infrastructure cost • diversity of lot sizes and configurations can potentially result in a diversity of housing styles and sizes
Construction Costs	• average cost per acre (streets & pipes): 10 acre - $179,775 50 acre - $186,380 100 acre - $193,005	• average cost per acre (streets & pipes) 10 acre - $136,950 50 acre - $138,110 100 ac re - $135,393
Maintenance and renewal costs	• average of $164,720/acre for surface and underground works (2009 current dollars)	• average of $143,900/acre for surface and underground works (2009 current dollars)
Land drainage	• more direct connections required to regional service lines • Requires larger LDS as there is greater run-off due to additional impervious surface area	• less direct connections required to regional service lines
Assessable frontage	• less assessable frontage primarily due to corner lots resulting in less property tax revenue for the municipality	• 14% more assessable frontage resulting in more property tax revenue for the municipality
Land use efficiency[14]	• 19.32 persons/ac (44.49 persons/ha)	• 22.17 persons/ac (51.90 persons/ha)


THANK
YOU!